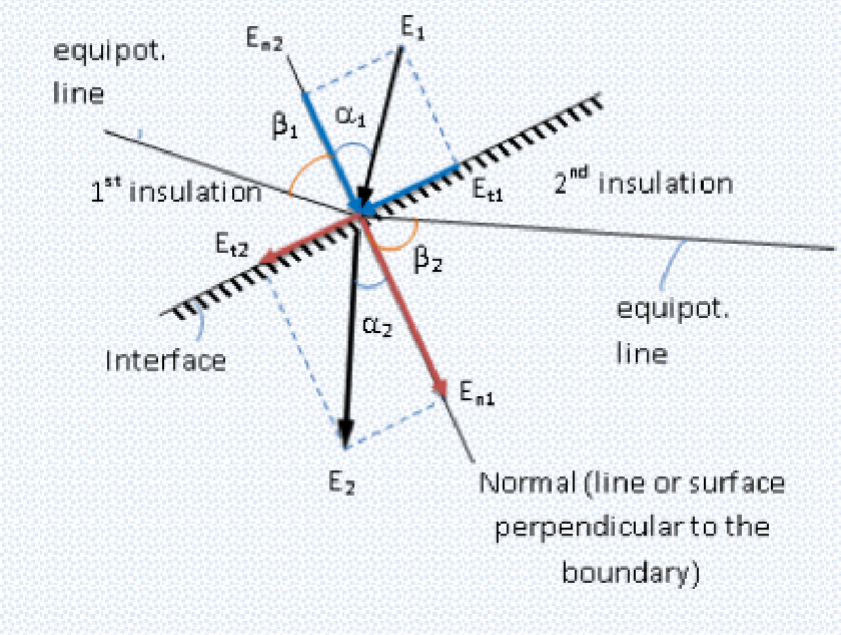
EE4047 – High Voltage Techniques I

# New Topic:

# Refraction of Electric Fields

## Week 13.1 – Monday Lesson




Asst. Prof. Dr. M. Alparslan

29.05.2023

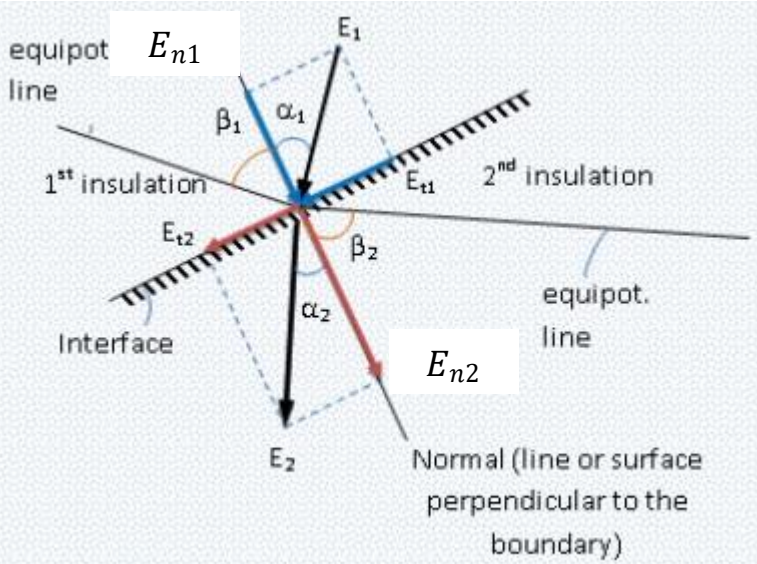
# OUTLINE

1 Introduction

2 Refraction

# 2 REFRACTION

## Refraction of electric field lines and equipotential lines




- $\alpha_1$, $\alpha_2$: Refraction angles of electric field lines.
- $\beta_1$, $\beta_2$: Refraction angles of equipotential lines

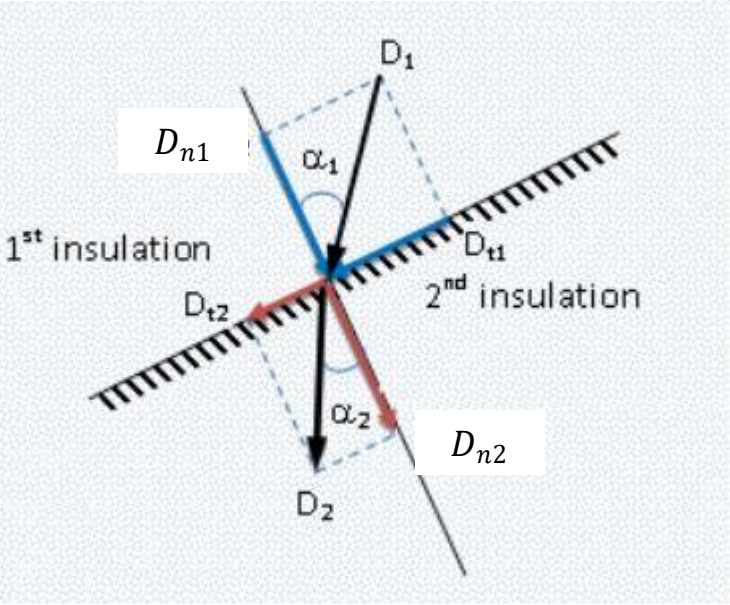
(REMEMBER: always perpendicular to the field lines!!!)

- $E_{t1}$, $E_{t2}$: tangent component of the electric fields $E_1$ and $E_2$
- $E_{n1}$, $E_{n2}$: normal (perpendicular) component of the electric fields $E_1$ and $E_2$

$$E_{t1} = E_{t2}$$

# 2 REFRACTION

## Refraction of electric field lines and equipotential lines

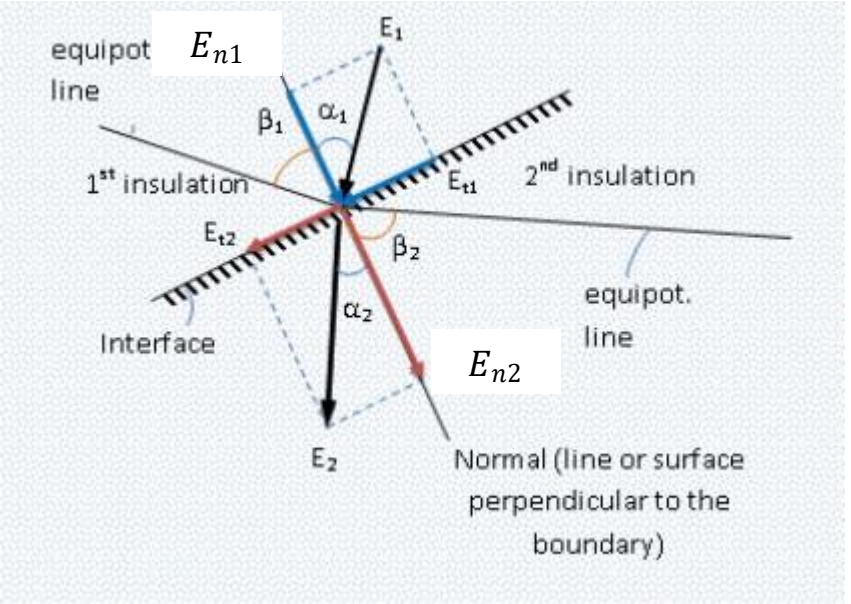



Similar to the electric field lines, displacement lines are refracted at the interface so that;

$$D_{n1} = D_{n2}$$

# 2 REFRACTION

## Simple trigonometric concepts




*For electric field vector components:*

$$E_{t1} = E_1 \sin \alpha_1$$
$$E_{t2} = E_2 \sin \alpha_2$$

*For displacement vector components:*

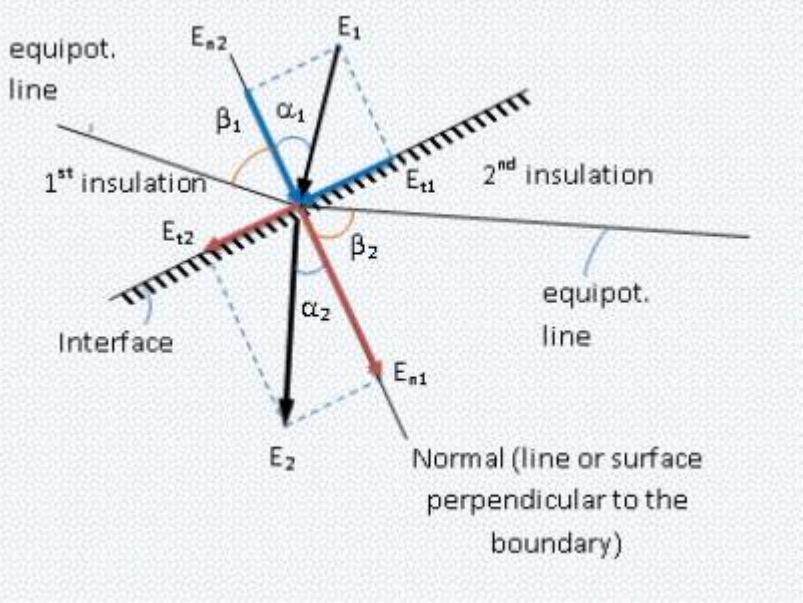
$$D_{n1} = D_1 \cos \alpha_1 = \varepsilon_1 E_1 \cos \alpha_1$$
$$D_{n2} = D_2 \cos \alpha_2 = \varepsilon_2 E_2 \cos \alpha_2$$

Because $E_{t1} = E_{t2}$ and $D_{n1} = D_{n2}$ ; we can write

$$\frac{E_{t1} = E_{t2}}{D_{n1} = D_{n2}}.$$

# 2 REFRACTION

## Refraction of equipotential lines




$$\beta_1 = 90 - \alpha_1$$
$$\beta_2 = 90 - \alpha_2$$

refraction angles of equipotential lines.

$$\frac{\tan(90-\beta_1)}{\tan(90-\beta_2)} = \frac{\varepsilon_1}{\varepsilon_2}$$

# 2 REFRACTION

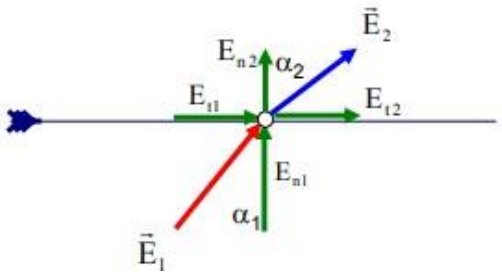
$$E_{t1} = E_{t2}$$

$$\frac{D_{t1}}{D_{t2}} = \frac{\varepsilon_1}{\varepsilon_2}$$

$$\vec{D} = \varepsilon.\vec{E}$$

$$\frac{E_{n1}}{E_{n2}} = \frac{\varepsilon_2}{\varepsilon_1}$$

$$D_{n1} = D_{n2}$$



# 2 REFRACTION

## Refraction of equipotential lines

$$E_{t1} = E_{t2} \Rightarrow |E_1|.\sin\alpha_1 = |E_2|.\sin\alpha_2 \qquad \frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2} \qquad \alpha_1 + \beta_1 = 90°$$

$$D_{n1} = D_{n2} \Rightarrow |D_1|.\cos\alpha_1 = |D_2|.\cos\alpha_2 \qquad \frac{\tan\beta_1}{\tan\beta_2} = \frac{\varepsilon_2}{\varepsilon_1} \qquad \alpha_2 + \beta_2 = 90°$$

$$|E_2| = \sqrt{E^2_{t2} + E^2_{n2}} = \sqrt{E^2_{t1} + (\frac{\varepsilon_1}{\varepsilon_2})^2.E^2_{n1}}$$

# 2 REFRACTION

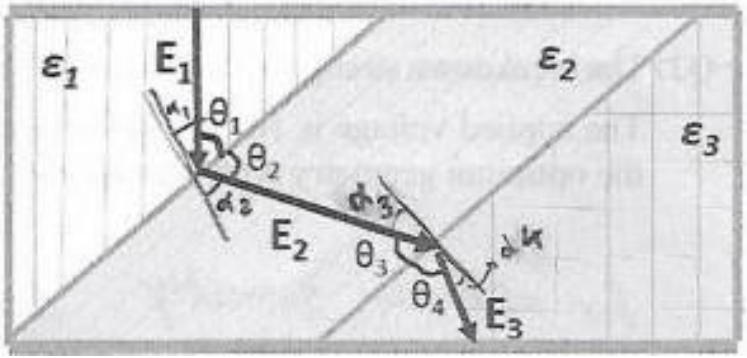
## Refraction of equipotential lines

Q4) In the following insulation system, there is a discharge in the 3rd insulator. The relative permittivity of the third insulator is $\varepsilon_{r3} = 1.2$ and its electric field strength is $E_{b3} = 100$ kV/cm. Suppose $\theta_1 = 45^\circ$, $\theta_2 = \theta_3 = 60^\circ$ and $\theta_4 = 75^\circ$. Determine:

a) The minimum values of $E_1$ and $E_2$.

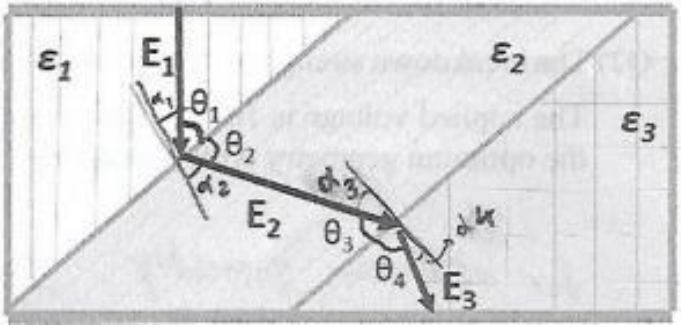
b) The relative permittivities of $\varepsilon_{r1}$ and $\varepsilon_{r2}$.

($\varepsilon_0 = 8.85 \times 10^{-12}$)



# 2 REFRACTION

## Refraction of equipotential lines




(a)

If breakdown happend in 3rd layer $\Rightarrow$ $E_3$ is minimum $100 \frac{kV}{cm}$

$$E_3 \sin\alpha_4 = E_2 \sin\alpha_3 \rightarrow 100 \sin(90^\circ - 75^\circ) = E_2 \sin(90^\circ - 60^\circ)$$

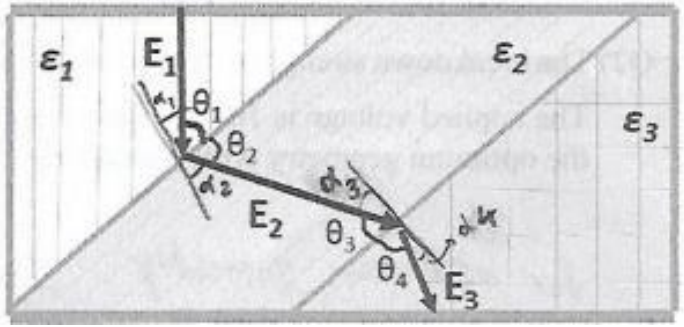
$$E_2 = \frac{100 \sin(15^\circ)}{\sin(30^\circ)} \rightarrow E_{2\,Minimum} = 51.76 \frac{kV}{cm}$$

$$E_2 \sin\alpha_2 = E_1 \sin\alpha_1 \longrightarrow 51.76 \sin(90^\circ - 60^\circ) = E_1 \sin(90^\circ - 45^\circ)$$

$$E_1 = \frac{51.76 \sin(30^\circ)}{\sin(45^\circ)} \Rightarrow E_{1\,Minimum} = 36.60 \frac{kV}{cm}$$

# 2 REFRACTION

## Refraction of equipotential lines




$$\frac{tan\alpha_2}{tan\alpha_3} = \frac{\varepsilon_2}{\varepsilon_3}$$

$$\frac{tan\alpha_2}{tan\alpha_3} = \frac{\varepsilon_{r2}}{\varepsilon_{r3}}$$

$$\frac{\tan(90-\theta_2)}{tan(90-\theta_3)} = \frac{\varepsilon_{r2}}{\varepsilon_{r3}}$$

(b)

$$\frac{\varepsilon_2}{\varepsilon_3} = \frac{tg\,\beta_2}{tg\,\beta_2} \rightarrow \frac{\varepsilon_0 \varepsilon_{r2}}{\varepsilon_0 (1.2)} = \frac{tg\,75^\circ}{tg\,60^\circ} \rightarrow \varepsilon_{r2} = 2.856$$

$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{tg\,\beta_2}{tg\,\beta_1} \rightarrow \frac{\varepsilon_0 \varepsilon_{r1}}{\varepsilon_0 (2.856)} = \frac{tg\,60^\circ}{tg\,45^\circ} \rightarrow \varepsilon_{r1} = 4.647$$

# 2 REFRACTION

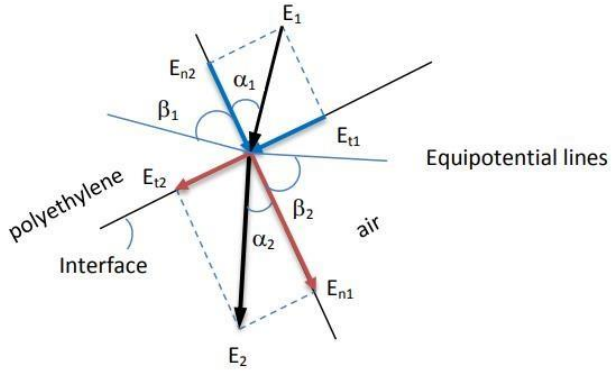
## Refraction of equipotential lines

**Q3.** An insulation environment consists of polyethylene ($\varepsilon_{r1} = 1$) and air ($\varepsilon_{r2} = 2.3$). Electric field vector comes to the interface with an angle of 55° through polyethylene.

**a.** Determine the refraction angles of electric field lines and equipotential lines.

**b.** If electric field strength in polyethylene is $E_1 = 90$ kV/cm, determine electric field strength in air, $E_2$; normal and tangential components of $E_1$ and $E_2$.

**c.** Does air have a breakdown considering breakdown strength of air is 30 kV/cm?



# 2 REFRACTION

## Refraction of equipotential lines

**a.** $E_1 = 90$ kV/cm

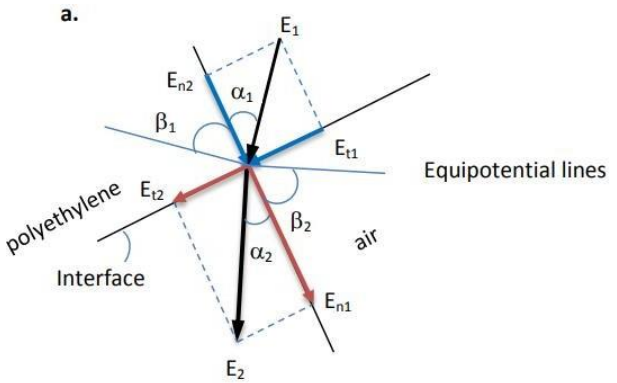
$\alpha_1 = 90 - 55 = 35°$

$\beta_1 = 90 - \alpha_1 = 55°$

$$\frac{\tan\alpha_1}{\tan\alpha_2} = \frac{\varepsilon_1}{\varepsilon_2} \rightarrow \tan\alpha_2 = \frac{1}{2.3}\tan 35°$$

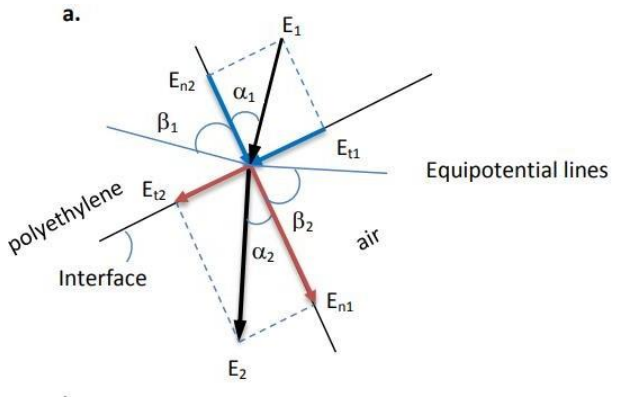
$\alpha_2 = 16.932°$

$\beta_2 = 90 - \alpha_2 = 73.068°$



# 2 REFRACTION

## Refraction of equipotential lines




b.

$$E_{t1} = E_1 \sin\alpha_1 = 90\frac{kV}{cm}\sin 35^\circ = 51.62\ \frac{kV}{cm}$$

$$E_{n1} = E_1\cos\alpha_1 = 90\frac{kV}{cm}\cos 35^\circ = 73.722\ \frac{kV}{cm}$$

$$E_{t2} = E_{t1} = E_2 \sin\alpha_2 \rightarrow E_2 = \frac{E_{t2}}{\sin\alpha_2} = \frac{51.62\ \frac{kV}{cm}}{\sin 16.932^\circ} = 177.244\frac{kV}{cm}$$

$$E_{n2} = E_2\cos\alpha_2 = 177.244\frac{kV}{cm}\cos 16.932^\circ{}^\circ = 169.56\ \frac{kV}{cm}$$

c. $E_{n2} = 169.56\ \frac{kV}{cm} > E_{b2} = 30\frac{kV}{cm} \rightarrow breakdown\ in\ air!$

E2 olacak yani 177.244>30

# END OF THE LESSON

## Any questions?

Check out the following video on Youtube:

Refraction of electric field lines: https://www.youtube.com/watch?v=A4VpnsKv5QI